

DecodeLabs Project 3 - SQL Data Analysis

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Database: week3_project

Project Objective

Perform SQL-based analysis using SELECT, WHERE, ORDER BY, GROUP BY, COUNT, SUM, and AVG operations on the sales dataset.

Record Count Verification

```
MySQL 8.0 Command Line Cli x + v
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.35 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE week3_project;
Database changed
mysql> SELECT COUNT(*) FROM sales;
+-----+
| COUNT(*) |
+-----+
|      1200 |
+-----+
1 row in set (0.00 sec)

mysql> |
```

SELECT Query Output

```
mysql> SELECT * FROM sales LIMIT 10;
+-----+-----+-----+-----+-----+-----+-----+-----+
| OrderID | Date       | CustomerID | Product | Quantity | UnitPrice | ShippingAddress | PaymentMeth
od | OrderStatus | TrackingNumber | ItemsInCart | CouponCode | ReferralSource | TotalPrice |
+-----+-----+-----+-----+-----+-----+-----+-----+
| ORD200000 | 2023-01-04 | C72649 | Monitor | 5 | 570.62 | 928 Main St | Debit Card
 | Shipped | TRK37947903 | 7 | SAVE10 | | Instagram | 2853.10 |
| ORD200001 | 2024-08-23 | C75739 | Phone | 2 | 151.35 | 823 Main St | Online
 | Shipped | TRK91186779 | 3 | SAVE10 | | Referral | 302.70 |
| ORD200002 | 2024-02-27 | C81728 | Tablet | 5 | 550.68 | 512 Main St | Credit Card
 | Cancelled | TRK42903982 | 8 | FREESHIP | | Email | 2753.40 |
| ORD200003 | 2023-10-15 | C33540 | Chair | 1 | 273.19 | 275 Main St | Debit Card
 | Returned | TRK62788070 | 5 | SAVE10 | | Facebook | 273.19 |
| ORD200004 | 2025-05-08 | C81840 | Printer | 4 | 626.01 | 668 Main St | Online
 | Delivered | TRK29241424 | 8 | SAVE10 | | Email | 2504.04 |
| ORD200005 | 2023-10-23 | C37249 | Phone | 2 | 245.86 | 934 Main St | Credit Card
 | Shipped | TRK72976927 | 4 | SAVE10 | | Instagram | 491.72 |
| ORD200006 | 2025-06-17 | C83492 | Laptop | 1 | 664.42 | 986 Main St | Gift Card
 | Returned | TRK96417362 | 6 | SAVE10 | | Facebook | 664.42 |
| ORD200007 | 2023-05-12 | C41460 | Monitor | 5 | 149.55 | 706 Main St | Cash
 | Shipped | TRK78809193 | 9 | FREESHIP | | Facebook | 747.75 |
| ORD200008 | 2025-04-02 | C26817 | Phone | 2 | 134.28 | 904 Main St | Gift Card
 | Cancelled | TRK61042692 | 2 | | | | Email | 268.56 |
| ORD200009 | 2023-11-21 | C31946 | Desk | 4 | 509.38 | 102 Main St | Credit Card
 | Shipped | TRK33478363 | 6 | SAVE10 | | Google | 2037.52 |
+-----+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> |
```

WHERE Clause Analysis

```
mysql> SELECT *
-> FROM sales
-> WHERE TotalPrice > 1000;
```

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.10
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.40
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
ORD200019	2023-01-23	C14552	Monitor	5	224.00	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120.00
ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6		Google	1371.80
ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093.00
ORD200029	2024-09-20	C32360	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRK59744090	6	SAVE10	Referral	1270.59
ORD200031	2023-06-30	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6		Facebook	1500.24
ORD200032	2023-05-01	C12388	Tablet	5	536.72	830 Main St	Cash	Delivered	TRK99261395	10		Email	2683.60
ORD200034	2023-01-07	C67195	Chair	4	576.52	926 Main St	Credit Card	Pending	TRK55467915	5	WINTER15	Facebook	2306.08
ORD200036	2025-05-08	C59337	Laptop	2	534.43	465 Main St	Debit Card	Pending	TRK50961250	5		Facebook	1068.86
ORD200041	2023-12-05	C20212	Monitor	3	611.45	657 Main St	Gift Card	Pending	TRK69733327	4	FREESHIP	Instagram	1834.35
ORD200043	2023-12-14	C60572	Chair	3	499.21	871 Main St	Online	Cancelled	TRK20351666	8		Instagram	1497.63
ORD200044	2025-04-21	C73225	Phone	2	550.18	996 Main St	Gift Card	Returned	TRK73048522	3	FREESHIP	Referral	1100.36
ORD200046	2024-06-26	C43212	Monitor	4	550.14	223 Main St	Debit Card	Pending	TRK25100762	4	FREESHIP	Instagram	2200.56
ORD200053	2024-05-18	C67551	Phone	4	488.78	512 Main St	Online	Returned	TRK94092552	5	FREESHIP	Google	1955.12

GROUP BY Product Count

```
mysql> SELECT *
-> FROM sales
-> ORDER BY TotalPrice DESC;
```

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200789	2023-08-17	C57276	Tablet	5	691.28	183 Main St	Online	Delivered	TRK75899752	10	SAVE10	Email	3456.40
ORD201122	2023-06-07	C38840	Monitor	5	678.19	766 Main St	Online	Returned	TRK32496970	8		Facebook	3390.95
ORD200632	2023-06-02	C67269	Laptop	5	670.16	463 Main St	Gift Card	Delivered	TRK82291044	7	WINTER15	Facebook	3390.00
ORD200469	2023-11-26	C13877	Chair	5	676.98	893 Main St	Cash	Cancelled	TRK17254691	5		Facebook	3384.90
ORD200328	2023-02-28	C18404	Tablet	5	674.04	546 Main St	Online	Cancelled	TRK89401624	7	SAVE10	Google	3370.20
ORD200107	2023-03-27	C16775	Printer	5	670.75	848 Main St	Gift Card	Shipped	TRK34392124	8	FREESHIP	Instagram	3353.75
ORD200326	2024-07-01	C65986	Laptop	5	670.48	273 Main St	Gift Card	Returned	TRK98353867	5	SAVE10	Facebook	3352.40
ORD201065	2023-10-30	C47778	Printer	5	666.80	408 Main St	Debit Card	Delivered	TRK73504329	7	SAVE10	Referral	3334.00
ORD201031	2023-02-28	C59183	Phone	5	664.51	136 Main St	Debit Card	Pending	TRK18129706	8	SAVE10	Email	3322.55
ORD200463	2023-05-26	C25276	Laptop	5	662.78	214 Main St	Debit Card	Shipped	TRK14011732	9	SAVE10	Instagram	3313.90
ORD200361	2024-06-29	C53464	Printer	5	659.85	197 Main St	Credit Card	Delivered	TRK32743531	9	SAVE10	Facebook	3299.25
ORD200367	2024-04-25	C13100	Chair	5	658.77	712 Main St	Gift Card	Pending	TRK83220400	10	FREESHIP	Instagram	3293.85
ORD200837	2025-06-14	C88029	Chair	5	656.85	282 Main St	Online	Delivered	TRK59701278	5		Email	3277.75
ORD200527	2024-08-19	C27202	Chair	5	653.47	967 Main St	Credit Card	Cancelled	TRK75406128	10	SAVE10	Referral	3267.35
ORD200768	2023-09-04	C35987	Tablet	5	653.46	165 Main St	Cash	Cancelled	TRK28918815	9	WINTER15	Email	3267.30
ORD200889	2024-10-27	C36725	Monitor	5	650.72	350 Main St	Credit Card	Cancelled	TRK16808790	8	WINTER15	Email	3253.60
ORD200540	2024-01-29	C87281	Laptop	5	648.65	852 Main St	Online	Pending	TRK81765057	7		Referral	3243.25
ORD200902	2024-02-17	C84331	Chair	5	644.64	205 Main St	Gift Card	Cancelled	TRK34440714	6	FREESHIP	Email	3223.20
ORD200957	2023-01-12	C73575	Monitor	5	643.89	851 Main St	Cash	Returned	TRK23387612	8	WINTER15	Facebook	3219.45
ORD200806	2024-06-19	C88205	Printer	5	643.03	447 Main St	Online	Cancelled	TRK93504166	5	FREESHIP	Referral	3215.15
ORD200221	2024-02-23	C93473	Tablet	5	639.37	552 Main St	Debit Card	Pending	TRK59003989	6	SAVE10	Google	3196.85
ORD200967	2025-05-06	C52079	Tablet	5	630.92	502 Main St	Credit Card	Returned	TRK67403204	5	FREESHIP	Google	3194.60
ORD200296	2024-06-19	C48453	Desk	5	638.80	434 Main St	Debit Card	Cancelled	TRK52110494	10	WINTER15	Referral	3194.00
ORD201079	2024-06-20	C89979	Phone	5	634.00	144 Main St	Gift Card	Delivered	TRK99085909	8		Email	3170.00
ORD200364	2023-11-24	C85282	Phone	5	628.74	740 Main St	Cash	Cancelled	TRK86767206	9	FREESHIP	Facebook	3143.70
ORD200764	2020-03-18	C35083	Laptop	5	627.43	806 Main St	Credit Card	Cancelled	TRK14016163	10	FREESHIP	Email	3137.15
ORD200010	2023-12-29	C43443	Tablet	5	626.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200820	2023-05-27	C35377	Phone	5	623.81	171 Main St	Cash	Returned	TRK17342157	6	FREESHIP	Email	3119.05
ORD200907	2025-01-21	C15624	Tablet	5	619.52	922 Main St	Debit Card	Pending	TRK40621141	9		Email	3097.60
ORD200241	2023-05-16	C20817	Chair	5	615.67	407 Main St	Online	Pending	TRK44164590	6	SAVE10	Email	3078.35
ORD200450	2024-01-15	C36408	Monitor	5	615.10	160 Main St	Gift Card	Returned	TRK90023253	10	WINTER15	Instagram	3075.50
ORD200876	2024-06-09	C90363	Chair	5	608.91	200 Main St	Gift Card	Pending	TRK21534173	6	SAVE10	Google	3044.55
ORD200492	2023-10-25	C39074	Laptop	5	606.52	103 Main St	Debit Card	Shipped	TRK13027036	6	SAVE10	Instagram	3032.60
ORD200633	2024-04-18	C79533	Laptop	5	601.72	921 Main St	Gift Card	Cancelled	TRK54711422	5	WINTER15	Instagram	3008.60
ORD200731	2024-06-07	C71310	Printer	5	584.46	308 Main St	Credit Card	Returned	TRK48292586	8		Email	2922.30
ORD200916	2024-06-21	C35669	Printer	5	578.83	165 Main St	Cash	Returned	TRK13920455	6		Referral	2894.15
ORD200511	2024-03-15	C11415	Monitor	5	575.24	267 Main St	Debit Card	Delivered	TRK65061457	7	WINTER15	Email	2876.20
ORD200816	2024-02-21	C52686	Tablet	5	574.79	497 Main St	Credit Card	Pending	TRK39085734	8	WINTER15	Referral	2873.95
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.10
ORD200870	2024-10-10	C50415	Monitor	5	566.07	587 Main St	Credit Card	Delivered	TRK41346534	6	SAVE10	Instagram	2830.35
ORD200683	2024-06-13	C43505	Printer	5	561.48	451 Main St	Online	Delivered	TRK25317340	9	FREESHIP	Facebook	2807.40
ORD201002	2025-06-19	C83911	Tablet	4	699.88	944 Main St	Cash	Pending	TRK33188106	5	WINTER15	Referral	2799.52
ORD200628	2023-03-26	C18717	Printer	5	558.63	861 Main St	Cash	Pending	TRK58568806	8	FREESHIP	Referral	2793.15
ORD200389	2023-12-17	C29703	Desk	4	697.50	558 Main St	Cash	Shipped	TRK40543461	7	FREESHIP	Referral	2790.00
ORD201153	2024-12-16	C98317	Monitor	4	696.71	276 Main St	Debit Card	Shipped	TRK37172138	4		Instagram	2786.04
ORD201142	2024-06-23	C23405	Phone	4	695.10	484 Main St	Debit Card	Returned	TRK83670708	8		Email	2780.40
ORD200222	2023-05-04	C43124	Chair	4	693.36	931 Main St	Online	Shipped	TRK17203338	8		Instagram	2773.44
ORD201087	2025-03-23	C84134	Laptop	4	693.07	530 Main St	Online	Shipped	TRK48659380	8	SAVE10	Instagram	2772.28
ORD200223	2024-09-11	C22783	Phone	4	692.91	673 Main St	Credit Card	Pending	TRK55009149	9	SAVE10	Instagram	2771.64
ORD201156	2023-07-19	C20512	Laptop	4	690.78	908 Main St	Credit Card	Shipped	TRK90374797	9		Referral	2763.12
ORD200471	2025-04-19	C52717	Desk	5	551.27	395 Main St	Online	Cancelled	TRK27625571	7	FREESHIP	Referral	2756.35
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.40

Revenue by Product

1200 rows in set (0.01 sec)

```
mysql> SELECT Product, COUNT(*) AS Orders  
-> FROM sales  
-> GROUP BY Product;
```

Product	Orders
Monitor	163
Phone	156
Tablet	179
Chair	178
Printer	181
Laptop	173
Desk	170

7 rows in set (0.01 sec)

```
mysql> |
```

Average Order Value

```
mysql> SELECT Product, SUM(TotalPrice) AS Revenue  
-> FROM sales  
-> GROUP BY Product;
```

Product	Revenue
Monitor	175651.41
Phone	151722.39
Tablet	186568.95
Chair	195620.11
Printer	195612.61
Laptop	192126.56
Desk	167459.93

7 rows in set (0.00 sec)

```
mysql>
```

Payment Method Analysis

```
mysql> SELECT AVG(TotalPrice) AS AvgOrderValue  
-> FROM sales;
```

AvgOrderValue
1053.968300

1 row in set (0.00 sec)

```
mysql> |
```

Order Status Analysis

```
mysql> SELECT PaymentMethod, COUNT(*) AS Total  
-> FROM sales  
-> GROUP BY PaymentMethod;
```

PaymentMethod	Total
Debit Card	232
Online	258
Credit Card	234
Gift Card	230
Cash	246

5 rows in set (0.00 sec)

Top Products by Revenue

```
mysql> SELECT OrderStatus, COUNT(*) AS Total
-> FROM sales
-> GROUP BY OrderStatus;
```

```
+-----+-----+
| OrderStatus | Total |
+-----+-----+
| Shipped     | 235   |
| Cancelled   | 250   |
| Returned    | 247   |
| Delivered   | 231   |
| Pending     | 237   |
+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> |
```

Additional Query Result

```
mysql> SELECT Product, SUM(TotalPrice) AS Revenue
-> FROM sales
-> GROUP BY Product
-> ORDER BY Revenue DESC
-> LIMIT 5;
```

```
+-----+-----+
| Product | Revenue |
+-----+-----+
| Chair   | 195620.11 |
| Printer | 195612.61 |
| Laptop  | 192126.56 |
| Tablet  | 186568.95 |
| Monitor | 175651.41 |
+-----+-----+
5 rows in set (0.00 sec)
```

Key Findings

1. Total records loaded: 1200
2. Chair generated the highest revenue.
3. Online payments were the most common payment method.
4. Order statuses were distributed across shipped, delivered, pending, cancelled and returned.
5. SQL concepts demonstrated: SELECT, WHERE, GROUP BY, ORDER BY, COUNT, SUM, AVG.